

AI Spring 2026

Quiz 1

Total Marks: 100

Date: 16/2/26

Cheating of any kind including use of ChatGPT etc will result in ZERO marks.

Q1: An autonomous **boring machine** is deployed on the surface of an asteroid to extract rare minerals. The robot uses thermal sensors and vibration detectors to identify mineral-rich veins while adjusting its drill speed to avoid overheating or causing structural collapses. It must navigate uneven terrain to return samples to a central collection lander.

a. Agent Type: Identify the agent type (simple reflex, model-based, goal-based, utility-based, learning). Justify based on the need to balance safety (overheating) vs. efficiency (collection).

b. Environment Classification: Classify the asteroid surface: Fully vs. partially observable? Deterministic vs. stochastic? Episodic vs. sequential? Static vs. dynamic? Discrete vs. continuous?

c. PEAS: Define the **Performance** measure, **Environment**, **Actuators**, and **Sensors** for this mining agent.

Q2: An AI must guess a 4-digit numeric password (0000–9999). It tests combinations sequentially without any additional information.

- What uninformed search strategy best models this situation?
- What is the branching factor and maximum depth of the search tree?
- Would Iterative Deepening Search be more efficient than DFS here? Explain.

Q3: An AI router must find the path with the lowest **latency** to send a data packet from **Server A** to **Gateway E**.

- Nodes:** Local Servers.
- Edges:** Fiber-optic links.
- Cost (g(n)):** Latency in milliseconds (ms).
- Heuristic (h(n)):** Estimated "hops" (remaining routers) to the destination.(Fiber Optic Links)

	A	B	C	D	E
A	0	4	5	∞	∞
B	4	0	∞	6	∞
C	5	∞	0	∞	4
D	∞	6	∞	0	2
E	∞	∞	4	2	0

- Using A*, compute **g(n)**, **h(n)**, and **f(n)** step-by-step until the goal is reached.
- What is the **optimal path** from A to E?
- Is the heuristic **admissible**? Justify briefly.